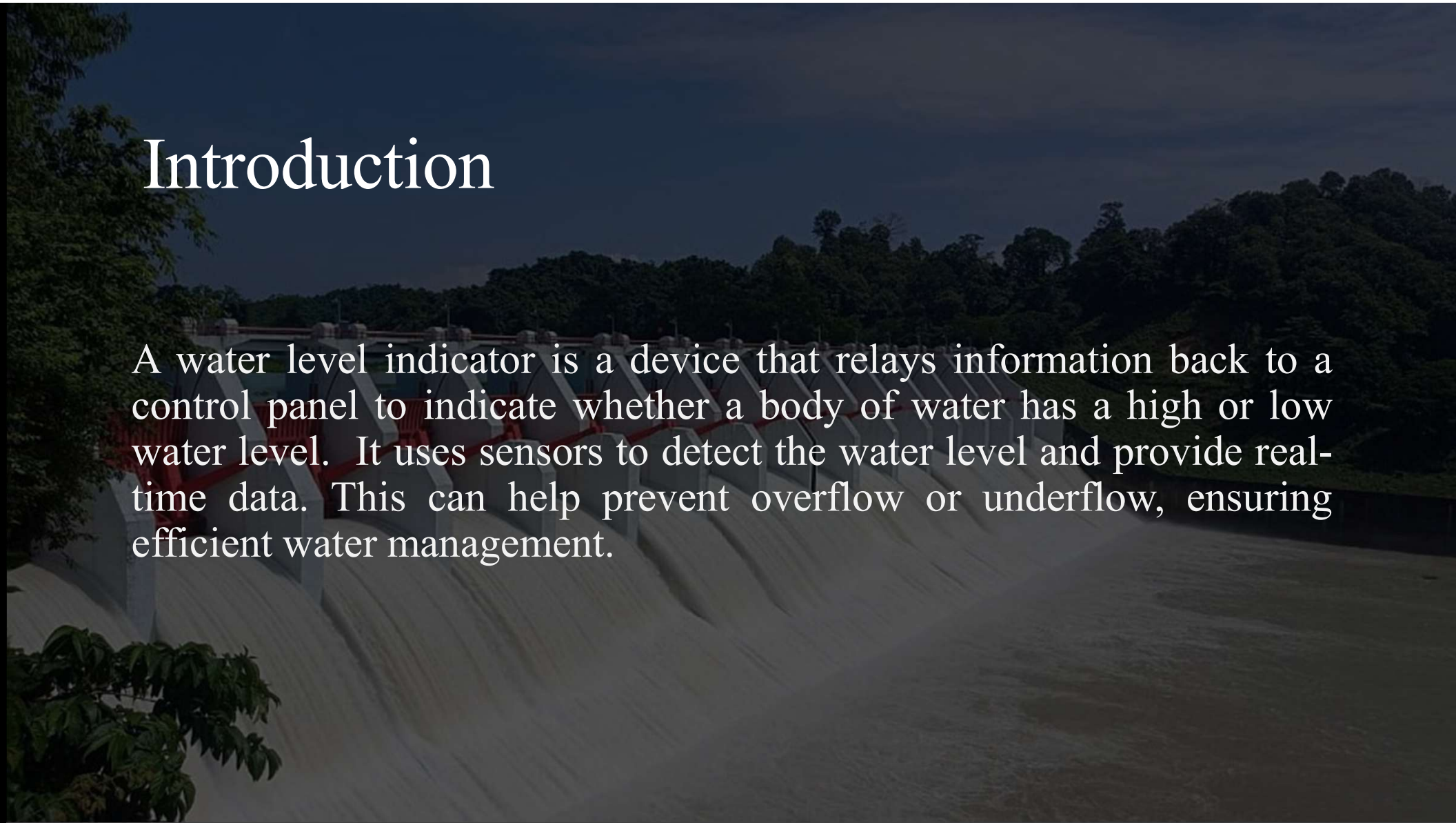


Water Level Indicator: Essential tools For monitoring Water resources

MONITORS AND DISPLAYS THE WATER LEVEL IN ANY RESERVOIR.

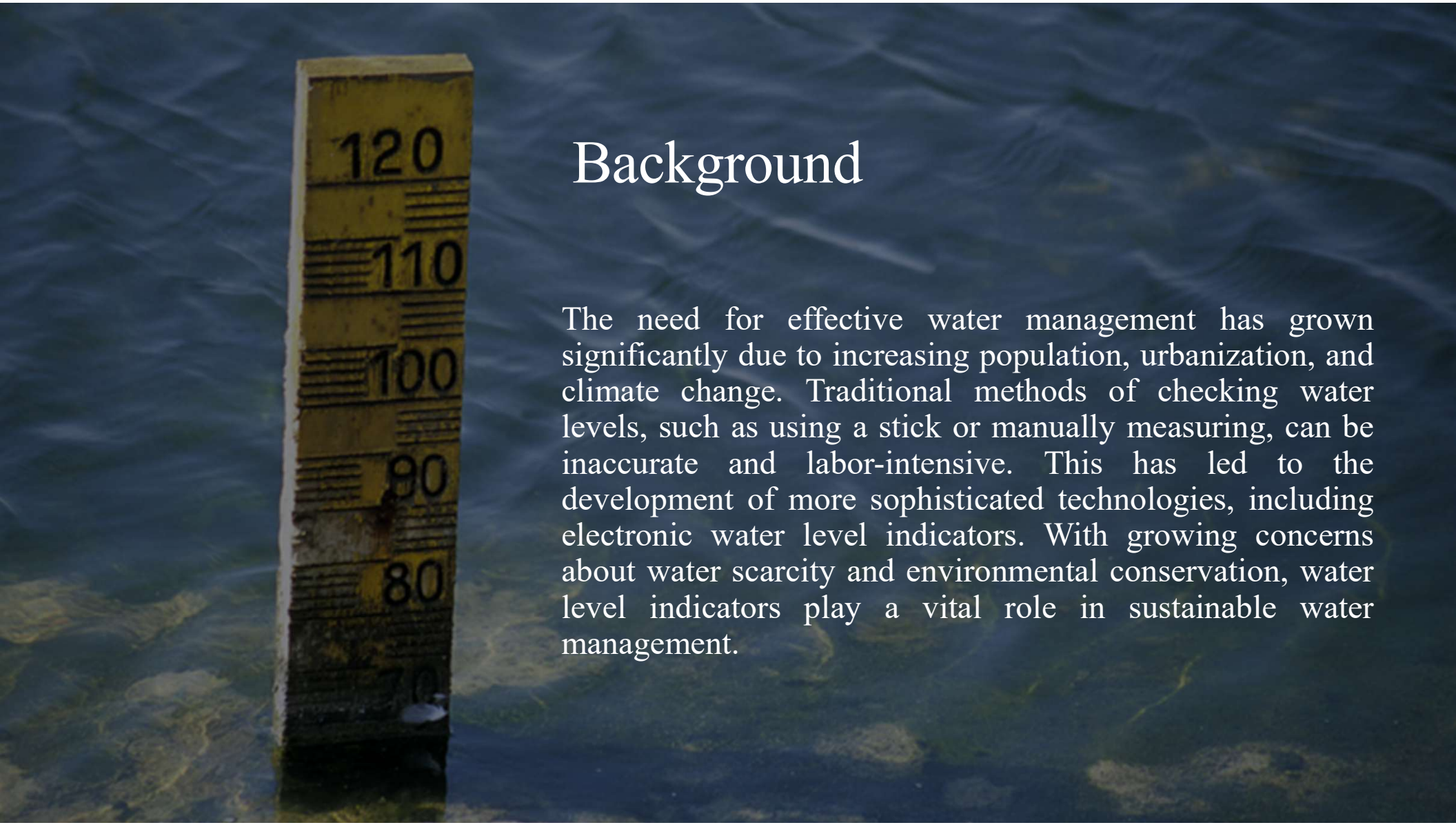
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Introduction

A water level indicator is a device that relays information back to a control panel to indicate whether a body of water has a high or low water level. It uses sensors to detect the water level and provide real-time data. This can help prevent overflow or underflow, ensuring efficient water management.



Background

The need for effective water management has grown significantly due to increasing population, urbanization, and climate change. Traditional methods of checking water levels, such as using a stick or manually measuring, can be inaccurate and labor-intensive. This has led to the development of more sophisticated technologies, including electronic water level indicators. With growing concerns about water scarcity and environmental conservation, water level indicators play a vital role in sustainable water management.

History



The concept of water level measurement has been around for centuries, dating back to ancient civilizations that used simple tools to measure water levels in rivers and lakes. With the advent of technology in the 20th century, mechanical and electronic systems were developed. Early water level Indicators used float switches, which would rise and fall with the water level, triggering alarms or pumps.



In recent years, advancements in sensor technology and microcontrollers have enabled the creation of more reliable and accurate water level indicators. These modern devices can provide real-time data, making water management more efficient and reducing waste.



This project focuses on designing a simple, effective water level indicator that can be used in various applications, improving water management practices in our communities.

Objectives of the Water Level Indicator Project



REAL-TIME MONITORING: TO CREATE A DEVICE THAT CONTINUOUSLY MONITORS WATER LEVELS



USER-FRIENDLY DESIGN: TO DEVELOP AN EASY INTERFACE.



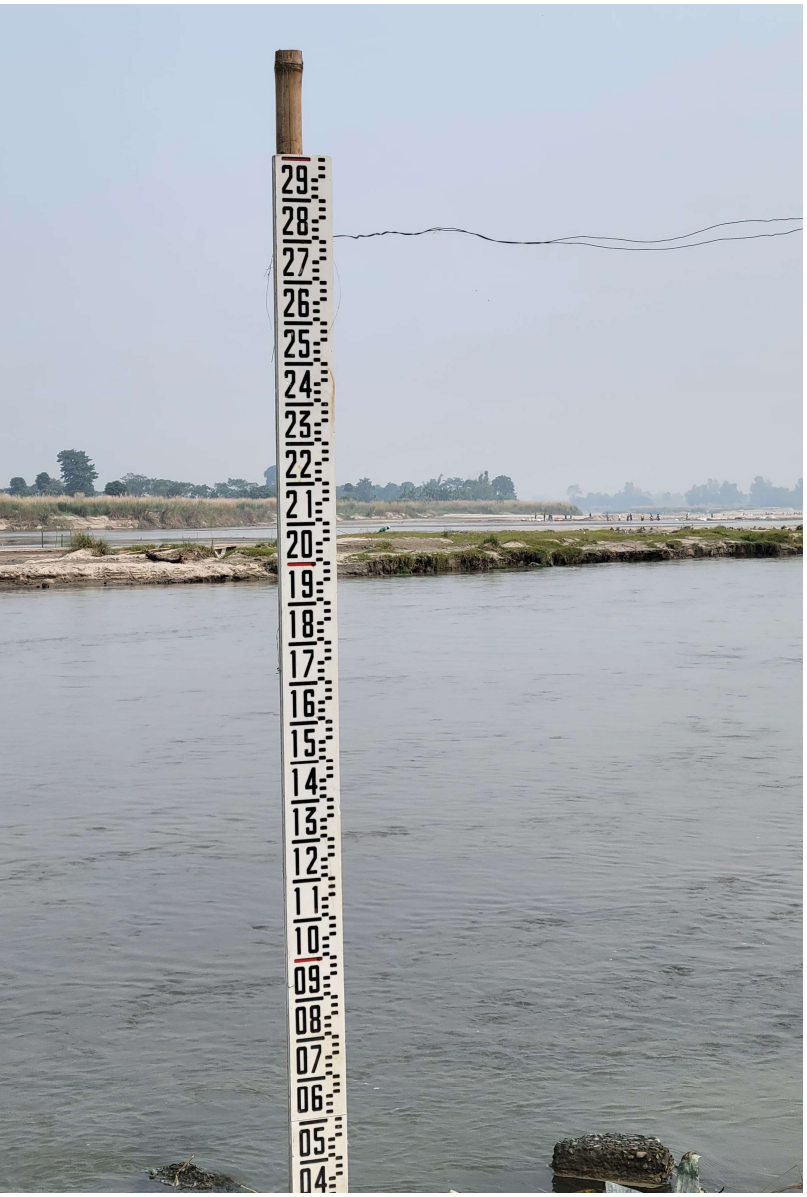
ACCURACY AND RELIABILITY: REDUCING ERRORS IN WATER MANAGEMENT.



ENERGY EFFICIENCY TO USE MINIMAL ENERGY, MAKING IT COST-EFFECTIVE AND ENVIRONMENTALLY FRIENDLY.



VERSATILE APPLICATIONS : TO MAKE THE WATER LEVEL INDICATOR SUITABLE FOR VARIOUS SETTINGS



Types Of Water Level Indicators:

1)Probe Sensors: These are placed at different levels in the tank. As the water level rises, it contacts the probes, sending signals to indicate the level.

2)Float Switches: These switches float on the water surface and move with the water level, triggering an action when the water reaches a certain point.

3)Ultrasonic Sensors: These measure the distance to the water surface using sound waves.

We are using Ultrasonic Sensor for measuring Water Level.

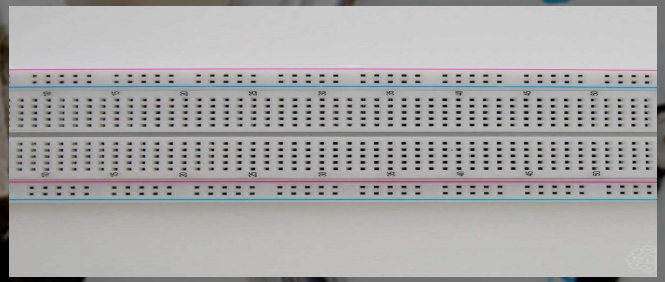
Key components of water level Indicators:

1. **Arduino UNO:** Arduino UNO is a low-cost, flexible, and easy-to-use programmable open-source microcontroller board that can be integrated into a variety of electronic projects. This board can be interfaced with other Arduino boards, Arduino shields, Raspberry Pi boards and can control relays, LEDs, servos, and motors as an output. This board contains a USB interface i.e. USB cable is used to connect the board with the computer and Arduino IDE (Integrated Development Environment) software is used to program the board.

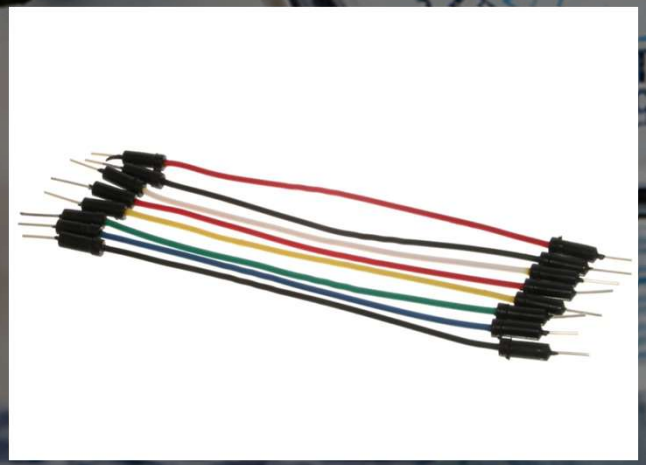


Key components of water level Indicators:

4. **Solderless breadboard**: A breadboard, solderless breadboard, or protoboard is a construction base used to build semi-permanent prototypes of electronic circuits.



5. **Jump wire**: A jump wire is an electrical wire, or group of them in a cable, with a connector or pin at each end , which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.

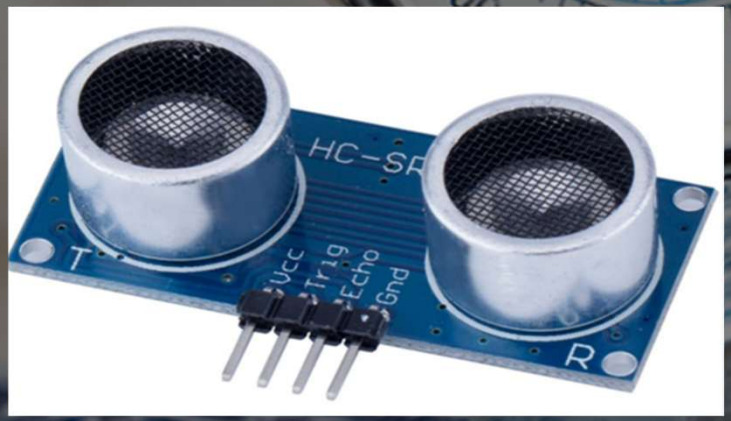


Key components of water level Indicators:

2**. LCD 12C** :An I2C LCD1602 is a device that can display text and characters on a 16x2 (16 columns and 2 rows) liquid crystal display (LCD) using the I2C protocol.



3.**Ultrasonic sensor**: An electronic instrument that uses ultrasonic sound waves (through air) to measure the distance of the target object and the reflected sound is converted into electrical signal. They are capable of detecting particles that are solid, liquid, granular, or powdered.



Key components of water level Indicators:

6. Pump : Pump is a mechanical device used to transfer different fluids from one position to another. Hydraulic device that lifts a fluid from a low to a high level and moves it from a low-pressure area to high pressure areas. Converting mechanical energy of the fluid into pressure energy.



Key components of water level Indicators:

7. **Battery** :A battery is a device that converts chemical energy contained within its active materials directly into electric energy by means of an electrochemical oxidation-reduction reaction. This type of reaction involves the transfer of electrons from one material to another via an electric circuit.



8. **Transparent pipe**: Clear plastic pipes, also referred to as transparent PVC pipes, are cylindrical tubes made from polyvinyl chloride (PVC) materials with transparent properties.



APPLICATION

Agriculture: Water level indicators monitor irrigation systems, optimizing water usage and boosting crop yields.

Buildings: Installed in tanks or reservoirs, indicators alert residents or owners to low water levels, preventing shortages.

Industry: Indicators in industrial tanks and equipment help prevent overflows, leaks, and ensure operational safety.

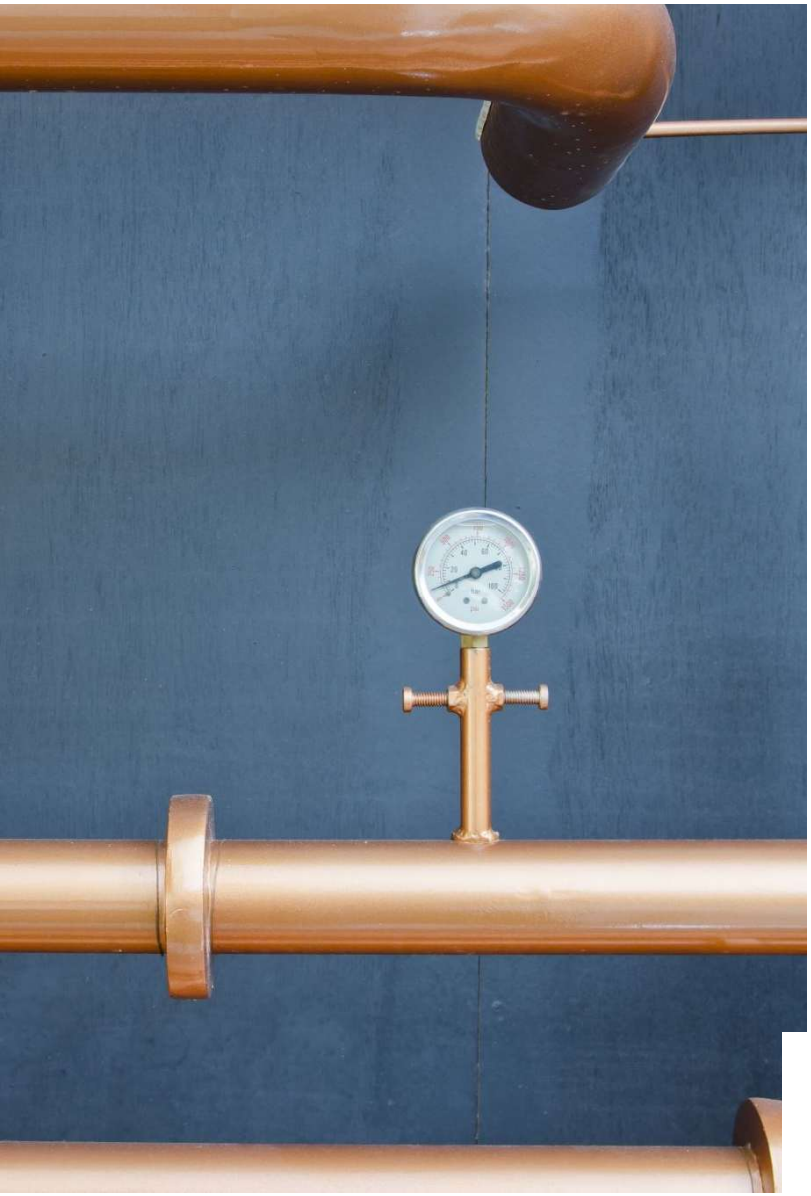
Environmental Monitoring: Used to track water levels in rivers and lakes, providing valuable data for flood risk assessment and climate impact studies.



APPLICATION

Home Water Tanks: Indicators monitor levels to prevent overflow.
Swimming Pools: Indicators help maintain proper water levels, supporting circulation and chemical balance.
Emergency Response: In flood systems, indicators provide early warnings to protect people and property in at-risk areas.





Methodology:

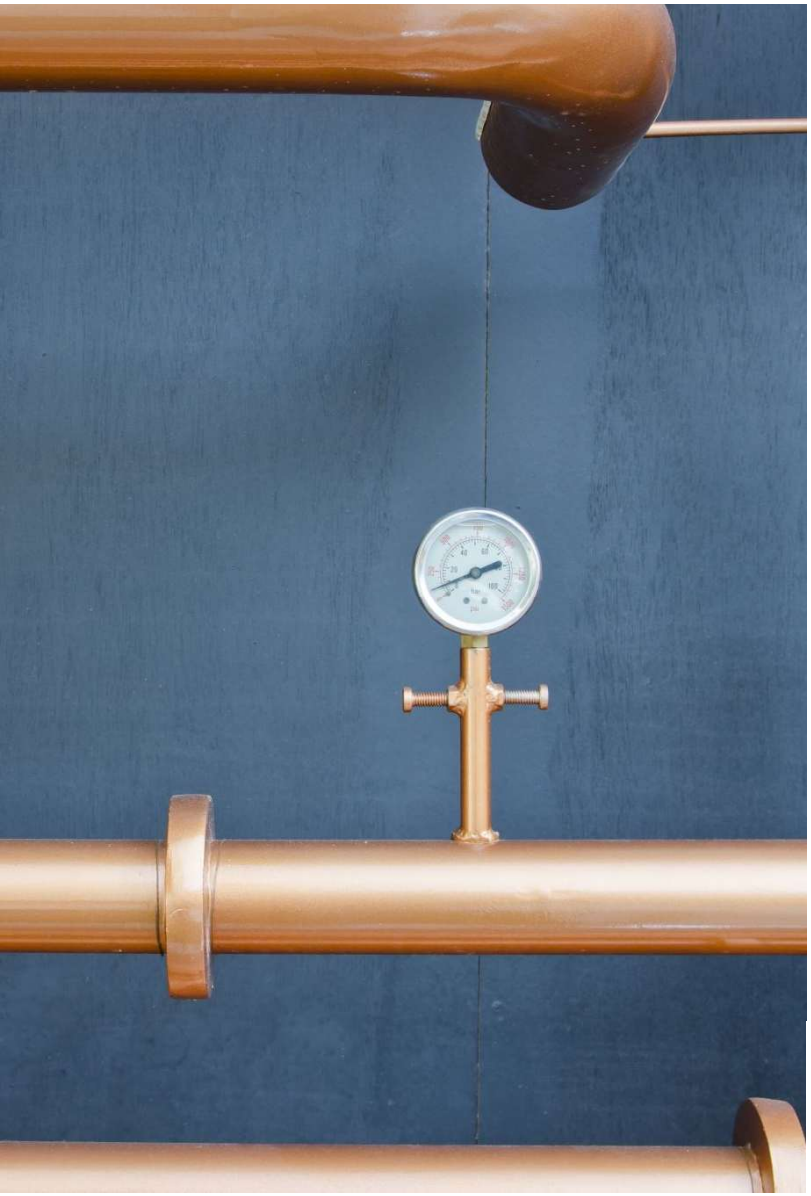
Design Phase: The key requirements for the water level monitoring system are defined, including tank size and the desired measurement accuracy. The appropriate type of sensor is selected based on the application.

Sensor Installation: Sensors are installed in optimal locations. For instance, if float switches are used, they are positioned at various heights within the tank to accurately detect different water levels. Each sensor is securely mounted and correctly positioned.

Microcontroller Programming: The microcontroller is programmed to read data from the sensors, with logic added to interpret the signals. For float switches, the program is set to recognize which water level each switch position represents.

Display Integration: A display is connected to the microcontroller and programmed to show the current water level based on sensor input, making monitoring straightforward.

Calibration: The system is tested with known water levels to calibrate sensor readings, ensuring accuracy and proper functionality.

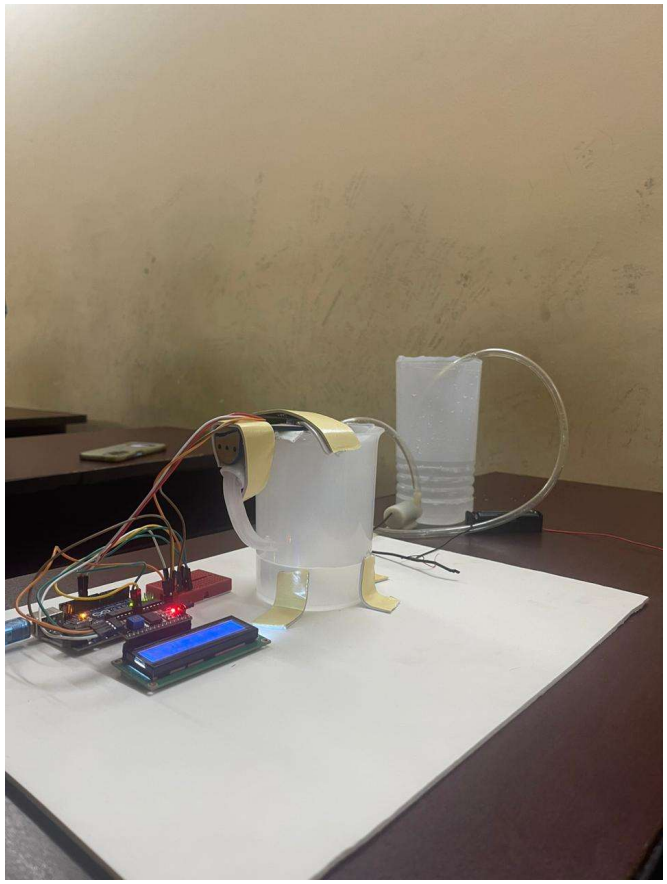


Methodology:

Testing : Conduct tests to ensure the system responds correctly to changing water levels . Check for any false readings or malfunctions.

Implementation : Install the complete system on the water tank . Ensure safety and reliability in the final installation.

Maintenance: Regularly check the system for proper functioning and calibrate sensors as needed.



Experimental Setup



Working Principle

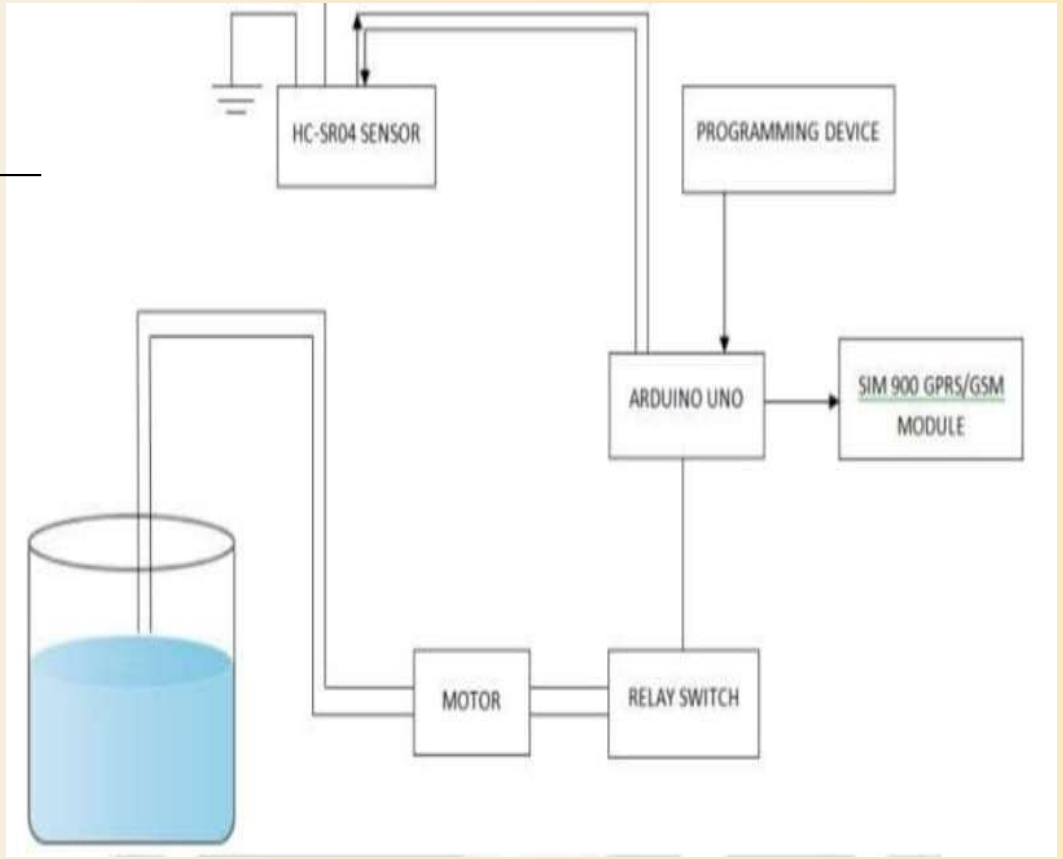
When the circuit is switched on the ultrasonic sensor transmits the generated sound signal to the bottom of the water tank which is the target and whose water level is to be measured. The signal after touching the base of the tank is reflected back and is received by the receiver of the ultrasonic sensor. The time taken through the entire journey of the transmitted signal is recorded. Then by applying the formula,

Range = {(time taken)*Velocity of the transmitted signal (i.e. 340 m/s)}/2

The output obtained is the required distance.

Working Principle

This measured level if is below the threshold level of 2cm then the pump will automatically be switched on and an SMS alert will be received at the user's phone. Henceforth, when the water reaches a particular level the motor will automatically turn off and again a notification through SMS will be provided indicating the tank is full to the user. This is the working principle used in our project. Let us see the the block diagram now



Code Display

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Pin assignments
const int trigPin = 12;   // TRIG pin
const int echoPin = 11;   // ECHO pin
const int buzzerPin = 4;  // Buzzer pin
const int relayPin = 7;   // Relay pin

// LCD I2C address (usually 0x27 or 0x3F)
LiquidCrystal_I2C lcd(0x27, 16, 2);  // Address 0x27, 16 columns, 2 rows

// Constants
const float tankHeight = 10.93; // Total height of the tank in cm
const float minWaterLevel = tankHeight * 0.25; // 25% of the tank height

const int numSamples = 10; // Number of samples for averaging

void setup() {
  // Initialize serial communication
  Serial.begin(9600);

  // Set pin modes
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, LOW);  // Ensure the pump is off at the start
```

```
| lcd.init();          // Initialize the LCD
  lcd.backlight();     // Turn on the backlight
  lcd.setCursor(0, 0); // Set the cursor to the first row
  lcd.print("Water Level Mon.");  // Initial message on the LCD
  delay(2000);         // Allow time for display
  lcd.clear();
}

// Function to take multiple sensor readings and return the average
float getAverageDistance() {
  float distances[numSamples];
  float sum = 0.0;

  for (int i = 0; i < numSamples; i++) {
    // Trigger the sensor
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    // Measure the echo duration
    long duration = pulseIn(echoPin, HIGH);

    // Calculate the distance in cm
    distances[i] = (duration * 0.0343) / 2;
    sum += distances[i];
```

```
  // Short delay between samples for stability
    delay(50) }
  // Calculate average distance
  float averageDistance = sum / numSamples;
  // Optional: Remove extreme outliers by calculating a new average
  float filteredSum = 0.0;
  int filteredCount = 0;
  for (int i = 0; i < numSamples; i++) {
    if (abs(distances[i] - averageDistance) < (0.1 * averageDistance)) { // 10% deviation threshold
      filteredSum += distances[i];
      filteredCount++;
    }
  }
  return filteredSum / filteredCount;
}
void loop() {
  float distance = getAverageDistance();
  // Calculate the water level (tankHeight - distance)
  float waterLevel = abs(tankHeight - distance);
  // Calculate the water level percentage
  float waterPercentage = abs(waterLevel / tankHeight) * 100.0;
  // Check if water level is below 25% of total height
  if (waterLevel < minWaterLevel) {
    // Turn on the buzzer and pump
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(relayPin, HIGH);
  } else {
    // Turn off the buzzer and pump
    digitalWrite(buzzerPin, LOW);
    digitalWrite(relayPin, LOW);
  }

  // Print the water level and percentage to the Serial Monitor
  Serial.print("Water Level: ");
  Serial.print(waterLevel, 2); // Two decimal places for precision
  Serial.println(" cm");

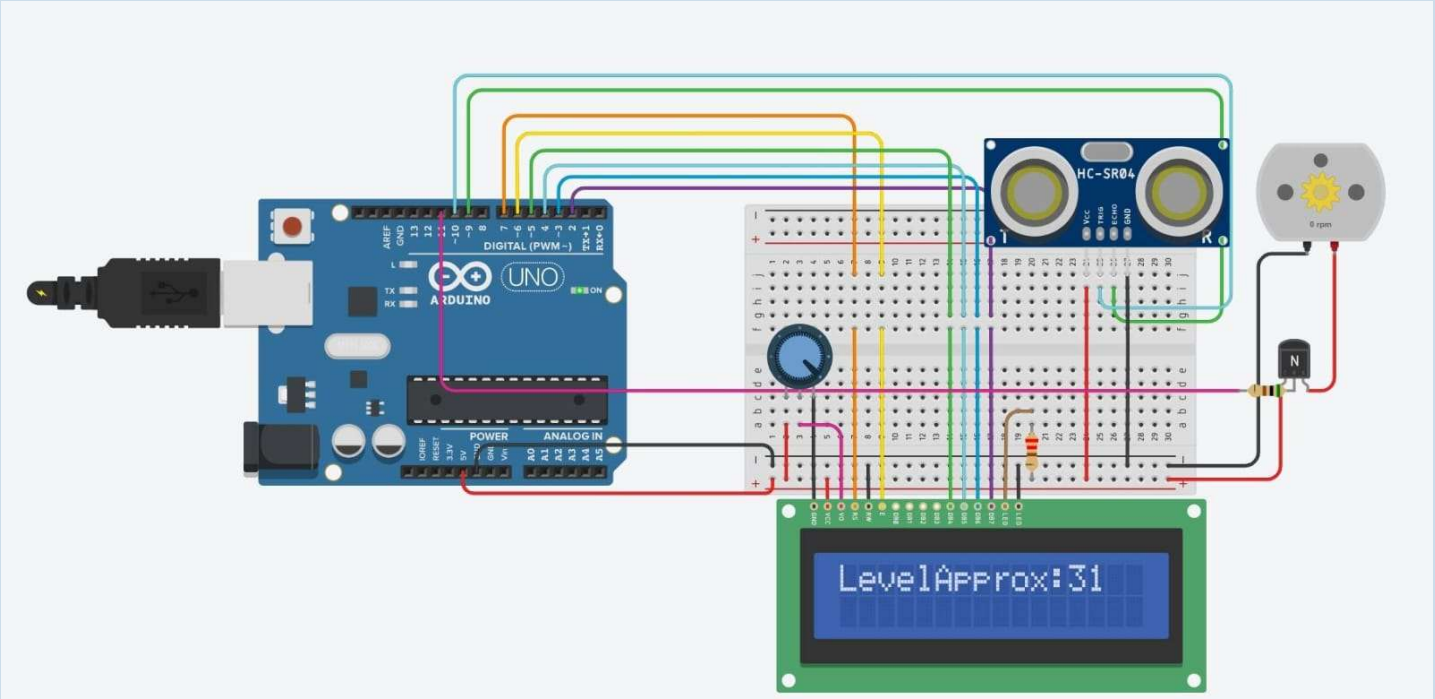
  Serial.print("Water Percentage: ");
  Serial.print(waterPercentage, 2);
  Serial.println(" %");

  // Display on LCD
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Level: ");
  lcd.print(waterLevel, 2);  // Display two decimal places
  lcd.print(" cm");

  lcd.setCursor(0, 1);
  lcd.print("Percent: ");
  lcd.print(waterPercentage, 2);  // Display two decimal places
  lcd.print(" %");

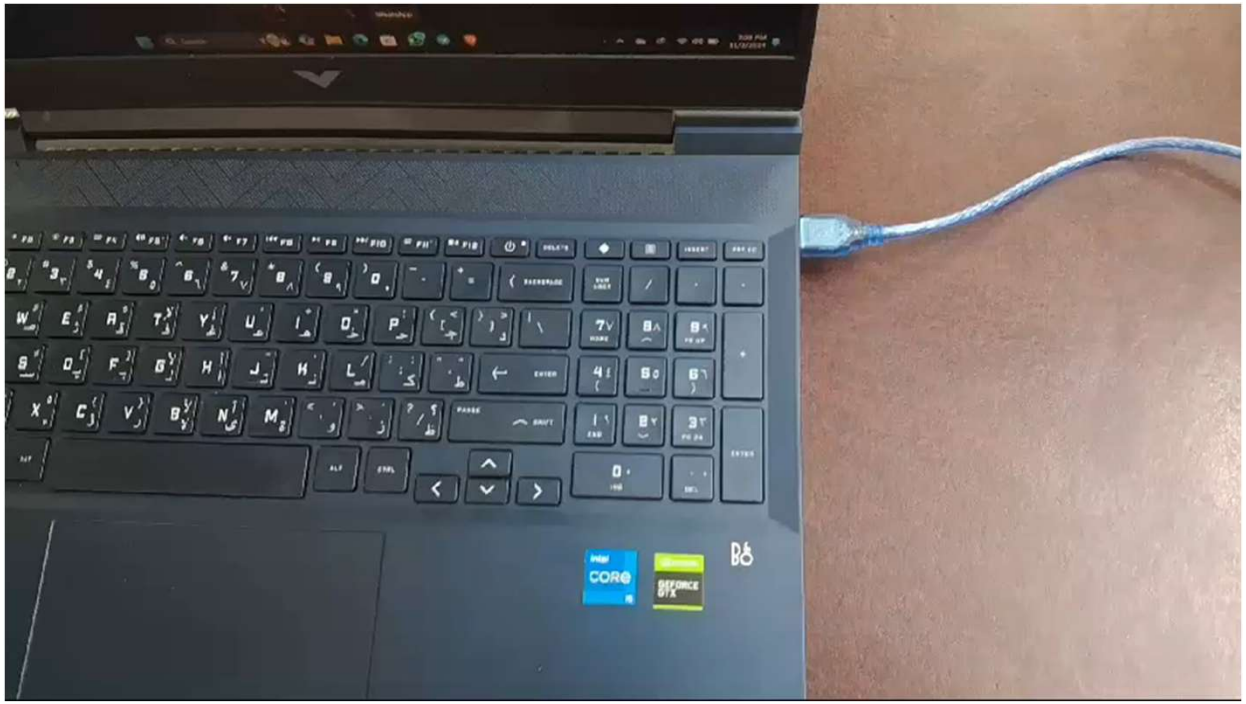
  // Delay before the next reading
  delay(1000);
}
```

Circuit Diagram:



Findings

1. The water level indicator project successfully helps in monitoring and indicating the water level in a tank or reservoir accurately.
2. The project provides a simple and cost-effective solution for checking the water level without having to physically inspect the tank.
3. The indicator system can be easily installed and maintained, making it suitable for both residential and commercial use.
4. The project can help in preventing water overflow or shortages by alerting users when the water level reaches a certain threshold.
5. The system is user-friendly and requires minimal technical expertise to operate effectively.
6. The water level indicator project can greatly help in water conservation efforts by promoting efficient water usage and management.


CORE
GEFORCE GTX

Conclusion

The project successfully demonstrates how technology can be leveraged for effective water management. By providing real-time water level monitoring and automated control, the system ensures efficient and sustainable water usage. The simplicity and affordability of the system make it a viable solution for various applications, contributing to better resource management and environmental conservation. Ultimately, the project underscores the importance of integrating technology into everyday life to solve practical problems and enhance efficiency. This project highlights the vital role of technological innovation in addressing the global challenge of water management. Through continuous improvements and adaptations, such systems can significantly contribute to sustainable development and resource conservation.

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4. Pandey, A., Andhale, G., Sonawane, A., Amrutkar, A., & Andhare, T. (2022). Automatic water level indicator and controller. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, *10*(11), 1043-1047.

Cost analysis for water level indicator

SL No.	Parts/Processes	Amount(Tk)
1	Arduino UNO	750
2	LCD 12C*2	360
3	Ultrasonic sensor*2	200
4	Solderless breadboard	40
5	Jump wire	100
6	Pump(5V)	100
7	Battery(3.5V)	100
8	Battery case	40
9	Transparent tube	20
10	Glue, cutter, tap	80
11	Project board	150
12	Project Jar	80
13	Modular	180
14	Buzzer*3	50
15	Sensor holder	50
16	Additional Cost	200
	Total	2500



Thank you!